In [1]:
```
import pandas as pd
import numpy as np
import sklearn
from sklearn import datasets
```

In [2]:
```
iris = datasets.load_iris()
iris
```

Out[2]:
```
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 'frame': None,
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0'),
 'DESCR': '.. _iris_dataset:\n\nIris plants dataset\n--------------------
-\n\n**Data Set Characteristics:**\n\n:Number of Instances: 150 (50 in e
ach of three classes)\n:Number of Attributes: 4 numeric, predictive attr
ibutes and the class\n:Attribute Information:\n    - sepal length in cm
\n    - sepal width in cm\n    - petal length in cm\n    - petal width i
n cm\n    - class:\n            - Iris-Setosa\n            - Iris-Versic
olour\n            - Iris-Virginica\n\n:Summary Statistics:\n\n=========
===== ==== ==== ======= ===== ====================\n                Min
Max   Mean    SD   Class Correlation\n============== ==== ==== ======= =
==== ====================\nsepal length:   4.3  7.9   5.84   0.83    0.7
826\nsepal width:    2.0  4.4   3.05   0.43   -0.4194\npetal length:
1.0  6.9   3.76   1.76    0.9490  (high!)\npetal width:    0.1  2.5   1.
20   0.76    0.9565  (high!)\n============== ==== ==== ======= ===== ===
=================\n\n:Missing Attribute Values: None\n:Class Distributio
```

```
n: 33.3% for each of 3 classes.\n:Creator: R.A. Fisher\n:Donor: Michael
Marshall (MARSHALL%PLU@io.arc.nasa.gov)\n:Date: July, 1988\n\nThe famous
Iris database, first used by Sir R.A. Fisher. The dataset is taken\nfrom
Fisher\'s paper. Note that it\'s the same as in R, but not as in the UCI
\nMachine Learning Repository, which has two wrong data points.\n\nThis
is perhaps the best known database to be found in the\npattern recogniti
on literature.  Fisher\'s paper is a classic in the field and\nis refere
nced frequently to this day.  (See Duda & Hart, for example.)  The\ndata
set contains 3 classes of 50 instances each, where each class refers to
a\ntype of iris plant.  One class is linearly separable from the other
2; the\nlatter are NOT linearly separable from each other.\n\n.. dropdow
n:: References\n\n  - Fisher, R.A. "The use of multiple measurements in
taxonomic problems"\n    Annual Eugenics, 7, Part II, 179-188 (1936); al
so in "Contributions to\n    Mathematical Statistics" (John Wiley, NY, 1
950).\n  - Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Sc
ene Analysis.\n    (Q327.D83) John Wiley & Sons.  ISBN 0-471-22361-1.  S
ee page 218.\n  - Dasarathy, B.V. (1980) "Nosing Around the Neighborhoo
d: A New System\n    Structure and Classification Rule for Recognition i
n Partially Exposed\n    Environments".  IEEE Transactions on Pattern An
alysis and Machine\n    Intelligence, Vol. PAMI-2, No. 1, 67-71.\n  - Ga
tes, G.W. (1972) "The Reduced Nearest Neighbor Rule".  IEEE Transactions
\n    on Information Theory, May 1972, 431-433.\n  - See also: 1988 MLC
Proceedings, 54-64.  Cheeseman et al"s AUTOCLASS II\n    conceptual clus
tering system finds 3 classes in the data.\n  - Many, many more ...\n',
 'feature_names': ['sepal length (cm)',
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 'filename': 'iris.csv',
 'data_module': 'sklearn.datasets.data'}
```

In [3]:
```
df = pd.DataFrame(iris['data'])
df.head()
```

Out[3]:

	0	1	2	3
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

In [4]:
```
df[4] = iris['target']
df.head()
```

Out[4]:

	0	1	2	3	4
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [5]:
```
df.rename(columns = {0:'SepalLengthCm', 1:'SepalWidthCm', 2:'PetalLengthC
df.head()
```

Out[5]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [6]:
```
df.describe()
```

Out[6]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

In [7]:
```
df.shape
```

Out[7]:
```
(150, 5)
```

MEAN

In [8]:
```
df.mean()
```

Out[8]:
```
SepalLengthCm    5.843333
SepalWidthCm     3.057333
PetalLengthCm    3.758000
PetalWidthCm     1.199333
Species          1.000000
dtype: float64
```

MEDIAN

In [9]:
```
df.median()
```

Out[9]:
```
SepalLengthCm    5.80
SepalWidthCm     3.00
PetalLengthCm    4.35
PetalWidthCm     1.30
Species          1.00
dtype: float64
```

MODE

In [10]:
```
df.Species.mode()
```

Out[10]:
```
0    0
1    1
2    2
Name: Species, dtype: int64
```

In [11]:
```
df.groupby(['Species']).count()
```

Out[11]:

Species	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	50	50	50	50
1	50	50	50	50
2	50	50	50	50

STANDARD DEVIATION

In [12]:
```
df.SepalLengthCm.std()
```

Out[12]:
```
np.float64(0.828066127977863)
```

In [13]:
```
df.SepalWidthCm.std()
```

Out[13]:
```
np.float64(0.4358662849366982)
```

In [14]:
```
df.PetalLengthCm.std()
```

Out[14]:
```
np.float64(1.7652982332594662)
```

In [15]:
```
df.PetalWidthCm.std()
```

Out[15]:
```
np.float64(0.7622376689603465)
```